

Kadathala Linganna, Seongmin Ju, Bokhyeon Kim, Won-Taek Han, Vemula Venkatramu and Chalicheemalapalli Kulala Jayasankar*

Spectroscopic Properties of Yb³⁺-Doped Silicate Glasses

DOI 10.1515/zpch-2017-0941

Received February 9, 2017; accepted April 16, 2017

Abstract: Yb³⁺-doped silicate glasses (SiO₂–Nb₂O₅–K₂O–ZnF₂) were prepared by a conventional melt quenching technique and their spectroscopic properties were investigated. The emission cross-section was found to be $0.53 \times 10^{-20} \text{ cm}^2$ at 1006 nm from the reciprocity method described by McCumber from the measured absorption spectrum of 1.0 mol% Yb₂O₃-doped glass. The lifetime (τ_{exp}) and figure of merit ($\sigma_{\text{em}} \times \tau_{\text{exp}}$) for the $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ transition of 1.0 mol% Yb₂O₃-doped silicate glass, were evaluated to be 0.50 ms and 0.26 cm² ms, respectively. The other spectroscopic parameters, β_{min} , I_{sat} and I_{min} were found to be 0.17, 29.00 kW/cm² and 4.90 kW/cm², respectively. The results indicate that the present Yb³⁺-doped silicate glasses could be useful as laser gain media at 1006 nm.

Keywords: absorption and emission cross-sections; lifetime; silicate glasses; Yb³⁺ ion.

1 Introduction

In recent years, the Yb³⁺-doped glasses have acquired a renewed interest for the development of ultra-short pulse and high energy diode pumped lasers because Yb³⁺ ion exhibits a broad emission bandwidth, a long excited-state lifetime, and a high saturation fluencies [1, 2]. Furthermore, the absorption spectrum of Yb³⁺ is located in the range of 900–1100 nm, therefore, the Yb³⁺-doped laser materi-

*Corresponding author: Chalicheemalapalli Kulala Jayasankar, Department of Physics, Sri Venkateswara University, Tirupati-517 502, India, e-mail: ckjaya@yahoo.com

Kadathala Linganna, Seongmin Ju and Won-Taek Han: School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology, 123 Cheomdangwagi-ro, Buk-gu, Gwangju 500-712, South Korea

Bokhyeon Kim: Advanced Photonics Research Institute, GIST, 1 Oryong-Dong, Buk-Gu, Gwangju 500-712, South Korea

Vemula Venkatramu: Department of Physics, Yogi Vemana University, Kadapa-516 003, India

als can be efficiently pumped by a low cost commercial available InGaAs laser diodes. Moreover, the low quantum defect of Yb^{3+} ion ensures low thermal effects and also its simple electronic structure eliminates detrimental processes such as quenching and excited state absorption [3–5]. For laser operation, an efficient host should show a combination of larger emission cross-section to provide high gain, longer fluorescence lifetime to minimize pump losses incurred from spontaneous emission, larger absorption cross-section at the pump wavelength, and the possibility to incorporate a high concentration of Yb^{3+} ions.

Among different glass hosts, silicate based glasses have been received great attention for solid-state laser media due to their special thermal and mechanical properties [6]. It is well known that inorganic glasses containing Nb_2O_5 are of considerable interest due to their promising physical and chemical properties. In particular, silicate glasses containing Nb_2O_5 contribute to the development of materials with higher non-linear optical coefficients, improved radiative properties, and laser glasses of high stimulated emission parameters [7]. On the other hand, addition of fluoride content to silicate glasses obviously improves the lifetimes and the emission cross-sections of Yb^{3+} ion.

Thus, the present work deals with the effect of Yb^{3+} concentration on the spectroscopic properties of the silicate glasses ($\text{SiO}_2\text{--Nb}_2\text{O}_5\text{--K}_2\text{O--ZnF}_2$). The absorption and emission cross-sections were evaluated from the absorption spectra using the reciprocity method. The decay curves for the excited ${}^2\text{F}_{5/2}$ level of Yb^{3+} were measured and lifetime was determined for all the studied glasses.

2 Experimental details

Yb^{3+} -doped silicate glasses with the composition (mol%) of $(40\text{--}x)\text{SiO}_2\text{--}20\text{Nb}_2\text{O}_5\text{--}20\text{K}_2\text{O--}20\text{ZnF}_2\text{--}x\text{Yb}_2\text{O}_3$ (where $x=0.01, 0.05, 0.1, 0.5$, and 1.0 , referred to as SNbKZF001, SNbKZF005, SNbKZF01, SNbKZF05, and SNbKZF10, respectively), were prepared by a standard melt quenching technique. About 20 g of the batch materials were thoroughly mixed/grinded in an agate mortar and the homogeneous mixture was taken in a platinum crucible and kept in an electric furnace at 1400°C for 3 h. The melt was then poured onto a preheated brass mold (400°C). After pouring, the glass samples were annealed at 550°C for 12 h and were allowed to cool slowly inside the furnace to room temperature in order to remove thermal strain and stress. Finally, the glass samples were cut and polished carefully for spectroscopic measurements.

The refractive index (n) was measured by a prism coupler method using laser source at 632.8 nm. The density was determined by Archimedes' method

using distilled water as an immersion liquid. Absorption spectra (300–1800 nm) were measured using a Carry series UV-Vis-NIR spectrophotometer. The near infrared emission (NIR) spectra was measured with a cw Ti:Sapphire laser (Mira-900, Coherent) at 808 nm associated with spectrophotometer (SpectraPro 2150i, Acton) and InGaAs detector (ID441, Acton). The decay curves were recorded by exciting the samples with laser diode at 980 nm and signal was detected with an InGaAs detector. All the measurements were carried out at room temperature.

3 Results and discussion

Figure 1 shows the absorption spectra of the Yb³⁺-doped silicate glasses for different Yb₂O₃ concentrations. It can be seen that the broad absorption in the range of 850–1100 nm of Yb³⁺ ions was observed due to the electronic transitions involving the Stark sublevels of the ²F_{7/2} and the ²F_{5/2} levels. The absorption peak at 976 nm appeared due to the 4f¹³–4f¹³ transitions between lower Stark sublevels of the ²F_{7/2} and ²F_{5/2} levels. It is also noticed that the absorption bandwidth and intensity increased with increasing Yb³⁺ concentration. The inset of Figure 1 shows the variation of absorption co-efficient at 976 nm with Yb³⁺ concentration. The observed increase in absorption co-efficient confirms the systematic incorporation of the Yb³⁺ ions within the glass matrix.

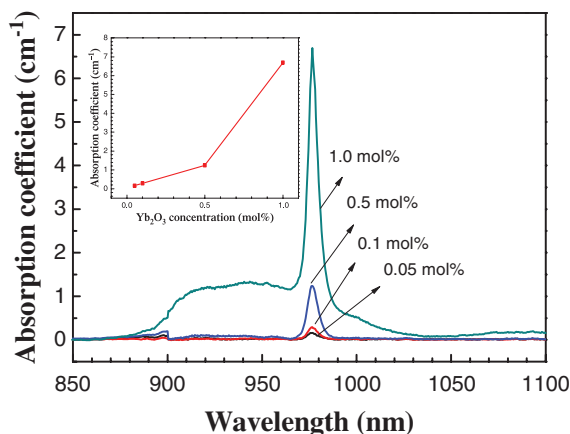


Fig. 1: Absorption coefficient spectra of Yb³⁺-doped silicate glasses. Inset shows the variation of absorption coefficient at 976 nm with Yb₂O₃ concentration.

The absorption cross-section (σ_{ab}) and the stimulated emission cross-section (σ_{em}) from the reciprocity method (RM) can be calculated by the following equations given by [8]

$$\sigma_{ab} = \frac{2.303 \log(I_0 / I)}{Nl} \quad (1)$$

$$\sigma_{em} = \sigma_{ab} \frac{Z_l}{Z_u} \exp \left[\frac{hc}{kT} \left(\frac{1}{\lambda_p} - \frac{1}{\lambda} \right) \right] \quad (2)$$

where N is the Yb^{3+} ion concentration (ions/cm³), l is the sample thickness, Z_l and Z_u represent the partition functions of the lower and upper states, respectively, λ_p is the peak absorption wavelength, h is the Planck's constant, c is the velocity of light, k is the Boltzman's constant, and T is the room temperature.

Figure 2 shows the absorption and emission cross-sections spectra of samples for different Yb_2O_3 concentrations. According to relationship between the integrated absorption cross-section and emission cross-section, it is evident that the increase of integrated absorption cross-section gives rise to an increase in the emission cross-section. As shown in Figure 2, the absorption and emission cross-sections increased with increasing Yb^{3+} concentration.

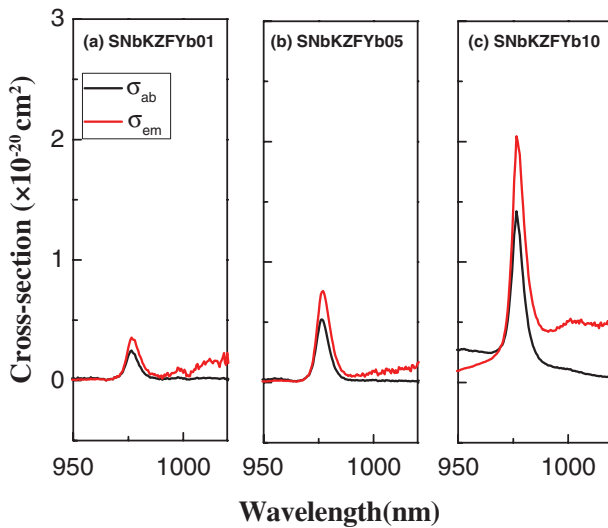


Fig. 2: Absorption and emission cross-section spectra of silicate glasses for different Yb^{3+} concentrations.

The σ_{ab} at 976 nm and σ_{em} at 1006 nm from RM, were evaluated for 1.0 mol% Yb₂O₃-doped silicate glass and are shown in Table 1 along with the previously reported glasses [6, 9–19] for comparison. It can be seen that the σ_{ab} was higher than the phosphate [9], lead phosphate [11], lead silicate [13], germanate [16], and silicate [18] glasses but lower than the calcium aluminosilicate [6], tellurite [10], fluorophosphate [12], bismuth borate [14], heavy metal oxide [15] and silicate [19] glasses. The σ_{em} at 1006 nm for the present SNbKZFb10 glass was comparable to the reported glasses [9, 11, 13–19] except for the calcium aluminosilicate [6], tellurite [10] and fluorophosphate [12] glasses.

Figure 3 shows the luminescence spectra of silicate glasses doped with various Yb³⁺ concentrations. As shown in Figure 3, a broad emission from 900 to 1150 nm has been observed for all the glasses and the intense emission peak located at 975 nm which corresponds to the transition between the lowest levels of ²F_{7/2} and ²F_{5/2} manifolds. Besides the emission at 975 nm, there is a broad band in the luminescence spectra peaked at 1006 nm, corresponding to the transitions mainly from the lowest level of ²F_{5/2} to the other levels in the ²F_{7/2} manifold except the lowest, and it is the favorable laser channel of Yb³⁺-doped laser materials. The intensity of the emission band increased when Yb³⁺ concentration was increased. The inset of Figure 3 shows the dependence of the integrated luminescence of the broad emission band with varying Yb³⁺ concentration. Additionally, no indication of emission near 500 nm which is a fingerprint of cooperative emission effects. Both facts furnish strong evidences that there is no cluster formation of Yb³⁺ in the present silicate glass system, which indicating this system is a good candidate for photonic applications or laser devices.

Figure 4 shows the decay curves for the ²F_{5/2} level for different Yb³⁺ concentrations. As can be seen, the decay curves exhibited a single exponential nature for all the glasses. The τ_{exp} of the ²F_{5/2} level of Yb³⁺ ion was determined using single exponential fitting and found to be 0.46, 0.51, 0.52, 0.55, and 0.50 ms for 0.01, 0.05, 0.1, 0.5, and 1.0 mol% of Yb₂O₃-doped glasses, respectively. It can be seen that the lifetime initially increased with increase in Yb₂O₃ concentration (≤ 0.5 mol%) and then decreased for further increase in Yb₂O₃ concentration (≥ 1.0 mol%). The initial rise in the lifetime was due to the radiative trapping or the radiative energy transfer between Yb³⁺ neighbor ions by ²F_{7/2} $\rightarrow$ ²F_{5/2} resonant transition [20–22]. In Yb³⁺ ion doped systems, the Yb³⁺ ion absorption and emission bands are superimposed at around 975 nm which causes to occur the detrimental radiation trapping process, where spontaneously emitted photons from excited state ²F_{5/2} excites the ²F_{7/2} ground state ions through re-absorption and leading to increase the lifetime of ²F_{5/2} level. The decreasing in lifetime for higher concentration due to self-quenching effect, which is associated to the enhancement of non-radiative energy transfer processes such as energy migration among Yb³⁺ ions followed

Tab. 1: Spectroscopic properties of Yb³⁺-doped glass systems.

Glass system	σ_{ab} (976 nm) (10 ⁻²⁰ cm ²)	σ_{em} (1006 nm) (10 ⁻²⁰ cm ²)	τ_{rad} (ms)	τ_{exp} (ms)	β_{min}	I_{sat} (kW/cm ²)	I_{min} (kW/cm ²)	$\sigma_{em} \times \tau_{exp}$ (10 ⁻²⁰ cm ² ms)
SNbKFYb10 [Present work]	1.42	0.53	0.47	0.50	0.17	29.00	4.90	0.26
Phosphate [9]	1.27	0.30	1.45	1.11	0.21	14.40	3.08	0.33
Tellurite [10]	1.93	1.21	0.37	0.63	—	—	2.74	0.77
Lead phosphate [11]	0.96	0.63	2.00	1.04	0.16	20.41	3.26	0.65
Fluorophosphate [12]	1.77	1.39	0.71	0.66	0.21	17	3.70	0.93
Lead silicate [13]	1.08	0.49	—	1.73	0.09	17	1.56	0.85
Bismuth borate [14]	2.44	0.93	0.43	0.39	0.19	21	4.07	0.36
Heavy metal oxide [15]	2.20	0.75	0.33	0.40	0.17	10	3.40	0.10
Germanate [16]	1.50	0.80	—	0.54	—	—	2.00	0.43
SBB30 [17]	—	0.54	—	1.04	0.11	22.54	2.47	0.56
SPb [17]	—	0.49	—	2.00	0.09	16.85	1.56	0.98
SB12 [17]	—	0.46	—	1.15	0.14	23.56	3.18	0.53
SAI [17]	—	0.32	—	1.42	0.16	22.78	3.76	0.45
Silicate [18]	0.85	0.48	—	1.86	—	—	—	—
Silicate [19]	2.40	0.77	—	0.78	—	—	—	0.60

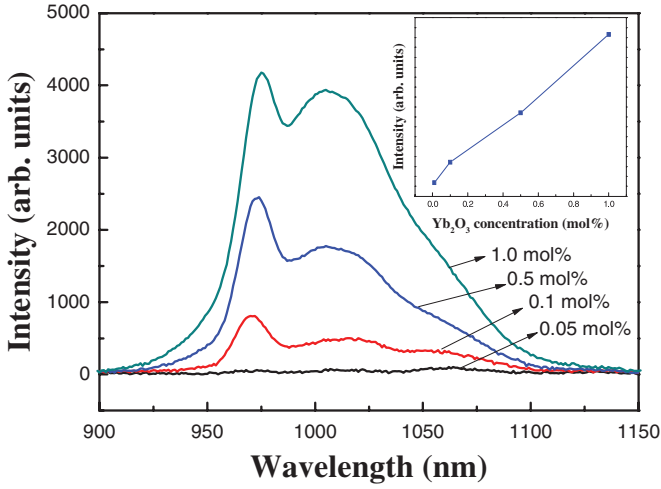


Fig. 3: Luminescence spectra of silicate glasses doped with various Yb³⁺ concentrations. Inset shows the integrated emission intensity as a function of Yb₂O₃ concentration.

by transfer of recombination centers and interaction between Yb³⁺ ions and the glassy host defects [16, 20, 22]. So, the optimum concentration was found to be 0.5 mol% in the present system. Table 1 presents the τ_{exp} for 1.0 mol% Yb₂O₃-doped silicate glass along with the reported glasses for comparison. It was noted that the τ_{exp} was comparable to the tellurite [10], fluorophosphate [12], bismuth borate [14], heavy metal oxide [15] and germanate [16] glasses except phosphate [9], lead phosphate [11] and silicate [6, 13, 17–19] glasses.

The spontaneous emission probability (A_R) can be evaluated using the following equation [23]:

$$A_R = \frac{8\pi cn^2(2J' + 1)}{\lambda_p^4 (2J + 1)N} \int k(\lambda) d\lambda \quad (3)$$

where 'c' represents the velocity of light, 'n' is the refractive index, ' λ_p ' is the absorption peak wavelength, 'N' is the concentration of Yb³⁺ ions, ' $k(\lambda)$ ' is the absorption coefficient, 'J' and 'J'' are the total momentum for the upper and lower levels. The A_R was calculated to be 2132 s⁻¹ using Eq. (3) and in turn used to obtain radiative lifetime (τ_{rad}). The τ_{rad} is shown in Table 1 for 1.0 mol% Yb₂O₃-doped silicate glass along with the reported glasses for comparison.

In order to assess the potential of the Yb³⁺-doped silicate glasses as a laser material, several important parameters such as the minimum pumping intensity (I_{min}), the minimum fraction of excited ions (β_{min}), and the saturation pumping

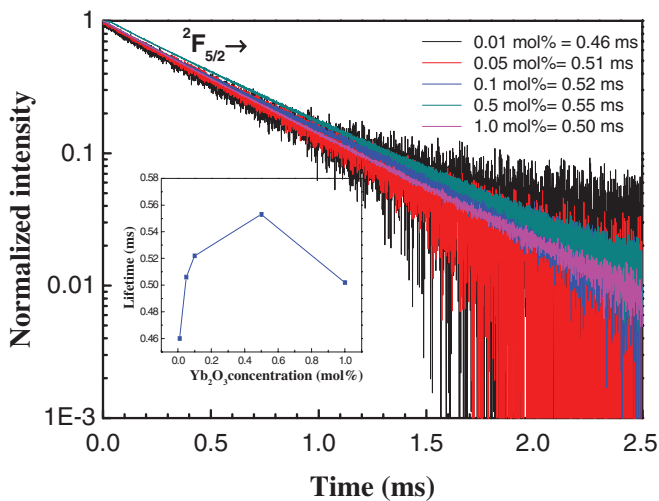


Fig. 4: Decay curves for the $^2F_{5/2}$ level of Yb^{3+} ion in silicate glasses. Inset shows the variation of lifetime with Yb_2O_3 concentration.

intensity (I_{sat}) were determined from the spectroscopic properties of absorption and emission cross-sections and experimental lifetime. The minimum absorbed pumping intensity (I_{min}), which is required for the transparency to be achieved at the lasing wavelength λ_0 (1006 nm), can be calculated by the following equations [23]:

$$I_{\text{min}} = \beta_{\text{min}} \times I_{\text{sat}} \quad (4)$$

where

$$\beta_{\text{min}} = \frac{\sigma_{\text{ab}}(\lambda_0)}{\sigma_{\text{em}}(\lambda_0) + \sigma_{\text{ab}}(\lambda_0)} = \left\{ 1 + \frac{Z_l}{Z_u} \exp \left[\frac{(E_l - hc\lambda_0)}{kT} \right] \right\}^{-1} \quad (5)$$

and

$$I_{\text{sat}} = \frac{hc}{\lambda_p \tau_{\text{exp}} \sigma_{\text{ab}}(\lambda_p)} \quad (6)$$

where $\sigma_{\text{ab}}(\lambda_0)$ and $\sigma_{\text{em}}(\lambda_0)$ represent the absorption and emission cross-section at the lasing wavelength λ_0 (1006 nm), λ_p , τ_{exp} and $\sigma_{\text{ab}}(\lambda_p)$ represent the wavelength of peak absorption (976 nm), fluorescence lifetime and absorption cross-section at the absorption wavelength, respectively. The parameters such as β_{min} , I_{sat} and I_{min} were determined for 1.0 mol% Yb_2O_3 -doped silicate glass and tabulated in Table 1 for comparison.

4 Conclusions

In this work, spectroscopic properties of Yb³⁺-doped silicate glasses have been studied. The linear increase of absorption coefficient with increase of Yb₂O₃ concentration confirms the systematic incorporation of Yb³⁺ ions in the glass matrix. No cooperative luminescence was found for all the concentrations, which an evidence of absence of clusters in the studied glasses. The broad emission bands peaked at 975 and 1006 nm were observed for all the samples and it was found that the integrated luminescence intensity increased with increasing Yb₂O₃ concentration. The absorption cross-section ($^2F_{7/2} \rightarrow ^2F_{5/2}$) and emission cross-section ($^2F_{5/2} \rightarrow ^2F_{7/2}$) were evaluated using the McCumber theory and found that they increased with increasing Yb₂O₃ concentration. The optimum Yb₂O₃ concentration was found to be 0.5 mol%, in the present glass system as evidenced from the lifetime variation. The figure of merit ($\sigma_{em} \times \tau_{exp}$) and other spectroscopic parameters, β_{min} , τ_{sat} and I_{min} were evaluated for the 1.0 mol% Yb³⁺-doped silicate glass. The results resemble that the Yb³⁺-doped silicate glasses might be useful as laser gain media around 1.0 μ m.

Acknowledgment: This work was also partially supported by the New Growth Engine Industry Project of the Ministry of Trade, Industry and Energy, Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2013R1A1A2063250), and KEPKO Research Institute (KEPRI) and managed by KESRI (Project Number: KEPRI-16-23), South Korea. One of the authors (C.K. Jayasankar) is grateful to DAE-BRNS for the award of Mega Research Project (No. 2009/34/36/BRNS/3174, dated: 12-02-2010).

References

1. M. Siebold, J. Hein, M. Hornung, S. Podleska, M. Kaluza, S. Bock, R. Sauerbrey, *Appl. Phys. B* **90** (2008) 431.
2. M. Siebold, F. Roeser, M. Loeser, D. Albach, U. Schramm, *Proc. SPIE* **8780** (2013) 878005.
3. B. J. Ainslie, S. P. Craig-Ryan, S. T. Davery, J. R. Armitage, C. G. Atkins, J. F. Massicot, R. Wyatt, *IEEE Proc.* **137** (1990) 205.
4. F. Auzel, D. Meichenin, F. Pelle, P. Goldner, *Opt. Mater.* **4** (1994) 35.
5. A. Malinowski, A. Piper, J. H. V. Price, K. Furusawa, Y. Jeong, J. Nilsson, D. J. Richardson, *Opt. Lett.* **29** (2004) 2073.
6. Y. Guyot, A. Steimacher, M. P. Belançon, A. N. Medina, M. L. Baesso, S. M. Lima, L. H. C. Andrade, A. Brenier, A.-M. Jurduc, G. Boulon, *J. Opt. Soc. Am. B* **28** (2011) 2510.
7. B. Samunova, V. Dimitrov, *J. Non-Cryst. Solids* **129** (1991) 54.
8. D. E. McCumber, *Phys. Rev.* **134** (1964) A299.

9. A. S. Pinheiro, A. M. Freitas, G. H. Silva, M. J. V. Bell, V. Anjos, A. P. Carmo, N. O. Dantas, *Chem. Phys. Lett.* **592** (2014) 164.
10. S. Balaji, A. D. Sontakke, K. Annapurna, *J. Opt. Soc. Am. B* **29** (2012) 1569.
11. K. Venkata Krishnaiah, R. Rajeswari, K. Upendra Kumar, S. Surendra Babu, I. R. Martín, C. K. Jayasankar, *J. Quant. Spectrosc. Radiat. Transf.* **140** (2014) 37.
12. J. H. Choi, A. Margaryan, A. Margaryan, F. G. Shi, *Mater. Res. Bull.* **40** (2005) 2189.
13. N. Dai, L. Hu, P. Lu, *Opt. Commun.* **253** (2015) 151.
14. Y. Chen, Y. Huang, Z. Luo, *Chem. Phys. Lett.* **382** (2003) 481.
15. L. R. P. Kassab, L. C. Courrol, A. S. Morais, C. M. S. P. Mendes, S. H. Tatumi, N. U. Wetter, L. Gomes, V. L. R. Salvador, *J. Non-Cryst. Solids* **304** (2002) 233.
16. L. R. P. Kassab, L. C. Courrol, V. D. D. Cacho, S. H. Tatumi, N. U. Wetter, L. Gomes, N. I. Morimoto, *J. Non-Cryst. Solids* **348** (2004) 103.
17. N. Dai, L. Hu, J. Yang, S. Dai, A. Lin, *J. Alloys Compd.* **363** (2004) 1.
18. N. Dai, L. Hu, W. Chen, G. Boulon, J. Yang, S. Dai, P. Lu, *J. Lumin.* **113** (2005) 221.
19. P. Barua, E. H. Sekya, K. Saito, A. J. Ikushima, *J. Non-Cryst. Solids* **354** (2008) 4760.
20. M. Ito, C. Goutaudier, Y. Guyot, K. Lebbou, T. Fukuda, G. Boulon, *J. Phys. Condens. Matter* **16** (2004) 1501.
21. S. Shen, M. Naftaly, A. Zha, *Opt. Commun.* **205** (2002) 101.
22. S. Xu, Z. Yang, S. Dai, J. Yang, L. Hu, Z. Jiang, *J. Alloys Compd.* **361** (2003) 313.
23. X. Zou, H. Toratani, *Phys. Rev. B* **52** (1995) 15889.